

1. Calculate the FT of $x(t)$ and given below using the linearity property.

$$x(t) = 2.e^{-5t}u(t-5) + 3.e^{-8|t|}$$

2. For given DT signal $x[n] = \cos(\omega_0 n)$
- Calculate the DTFS coefficients.
 - Calculate DTFT of for two different fundamental frequency ($\omega_{o1} = \frac{2\pi}{7}$ and $\omega_{o2} = \pi/2$). And sketch their DTFTs.
3. Calculate the FT of $y(t)$ given below and sketch the output in ω domain.

$$y(t) = \begin{cases} 1/2 & |t| < 5 \\ 0 & |t| > 5 \end{cases}$$

4. Calculate the inverse FT for following ω domain representations;
- $X(j\omega) = e^{-5\omega} u(\omega)$
 - $X(j\omega) = e^{-5|\omega|}$